

Module4_Lab1

September 24, 2026

```
[1]: import pandas as pd
df = pd.read_csv(sales_clean.csv)
df.shape
```

```
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NameError                                Traceback (most recent call last)
Cell In[1], line 2
      1 import pandas as pd
----> 2 df = pd.read_csv(sales_clean.csv)
      3 df.shape

NameError: name 'sales_clean' is not defined
```

```
[2]: import pandas as pd
df = pd.read_csv("sales_clean.csv")
df.shape
```

[2]: (292, 6)

```
[3]: print(df["price"].mean())
print(df["price"].median())
print(df["price"].std())
print(df["qty"].mean())
print(df["qty"].median())
print(df["qty"].std())
```

```
54.93099315068493
34.480000000000004
72.06522228697717
2.9863013698630136
3.0
1.4092785381929205
```

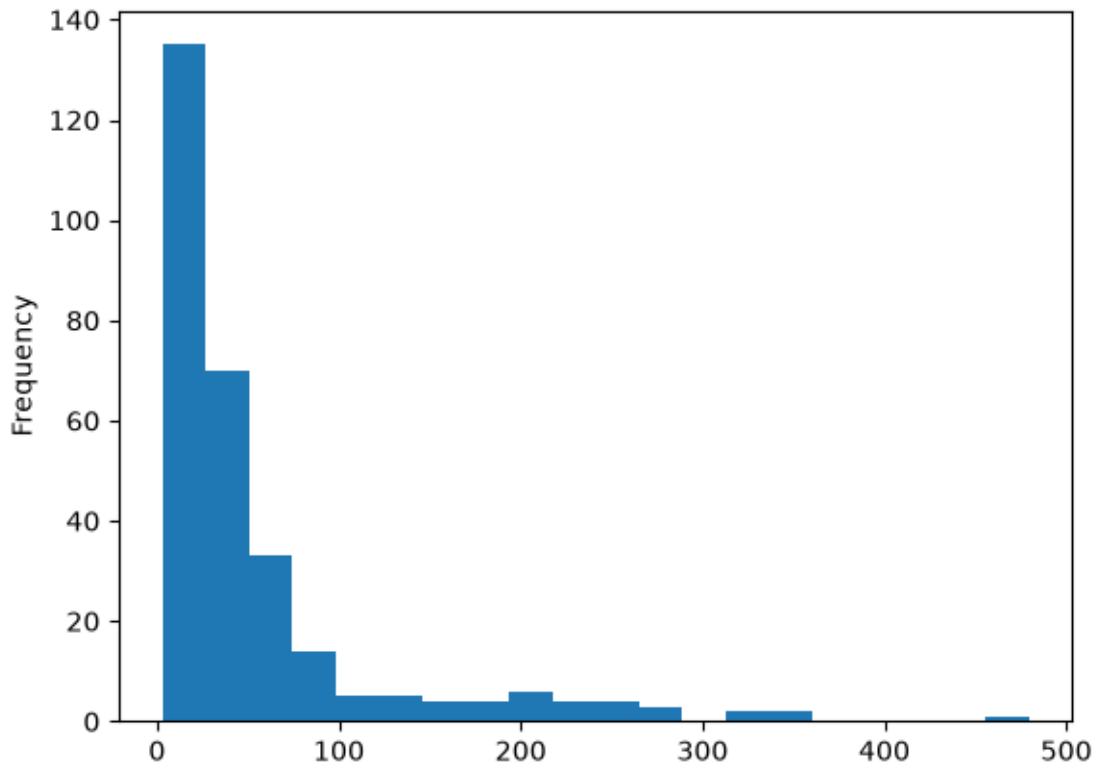
The mean is significantly higher than the median, so this indicates that the distribution is skewed to the right.

Which of mean or median would you report for price, and why? If the median price is \$34.48, that means exactly half of the items are below \$34.48. That may appeal to more buyers versus if we

were to report the mean of \$54.93 for price, it seems kind of expensive and you can probably only attract certain buyers who may be willing to spend a little more.

```
[4]: df["price"].plot(kind="hist", bins=20)
```

```
[4]: <Axes: ylabel='Frequency'>
```



This histogram shows that the distribution is skewed to the right

```
[5]: df["revenue"] = df["price"] * df["qty"]
```

```
[6]: df.head()
```

```
[6]:
```

	order_id	date	product	price	qty	zip	revenue
0	1254	2026-04-06	mug	11.36	4	60614	45.44
1	1057	2026-05-30	charger	14.79	3	30303	44.37
2	1150	2026-05-06	mug	5.50	1	10001	5.50
3	1066	2026-05-30	notebook	5.50	1	98101	5.50
4	1114	2026-04-06	webcam	45.59	4	90405	182.36

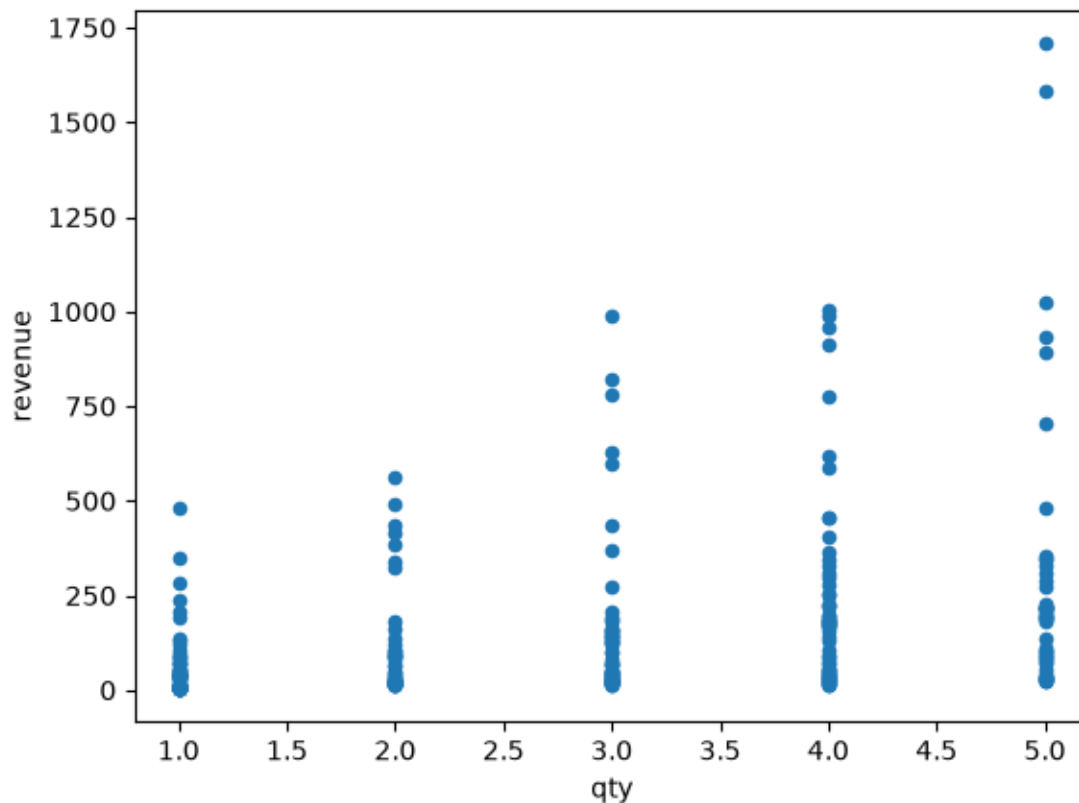
```
[7]: df.corr(numeric_only=True)
```

```
[7]:
```

	order_id	price	qty	zip	revenue
order_id	1.000000	-0.005043	0.031420	-0.014784	0.005373
price	-0.005043	1.000000	-0.034062	0.024285	0.822480
qty	0.031420	-0.034062	1.000000	0.086869	0.315427
zip	-0.014784	0.024285	0.086869	1.000000	0.048578
revenue	0.005373	0.822480	0.315427	0.048578	1.000000

```
[8]: df.plot.scatter(x="qty", y="revenue")
```

```
[8]: <Axes: xlabel='qty', ylabel='revenue'>
```



The scatter plot above tells us the number of items bought against the revenue. Majority of the dots are at the bottom, meaning most of the buyers buy lower priced items. As the quantity increases, the spread of the revenue expands as well.

Compute mean and median of [2, 4, 4, 6, 9] To compute mean, we take the sum of $2+4+4+6+9 = 25$, then divide $25 / 5 = 5$ To compute the median, we sort our group of numbers, which they already are. There are 5 values, and the number in the middle is the median, which is 4.

```
[9]: data = [2, 4, 4, 6, 9]
total = sum(data)
```

```
[10]: data = [2, 4, 4, 6, 9]
      total = sum(data)
      print(total)
```

25

```
[12]: a = 25
      b = 5
      result = a / b
      print(result)
```

5.0

Mean = 5.0

```
[15]: import statistics
      data = [2, 4, 4, 6, 9]
      median = statistics.median(data)
      print(median)
```

```
Cell In[15], line 4
      print(median))
              ^
```

SyntaxError: unmatched ')'

```
[16]: import statistics
      data = [2, 4, 4, 6, 9]
      median = statistics.median(data)
      print(median)
```

4

Median is 4